Lab – Working with Docker Container Commands

To Begin with the Lab

Summary

Docker provides several commands to manage containers through the Docker CLI (Command Line Interface). Although Docker Desktop provides a graphical interface to create, start, stop, and delete containers, it is important to understand the underlying Docker commands.

Using Docker commands, we can list running containers, view stopped containers, stop and restart existing containers, and remove containers when they are no longer required.

In this lab, we will learn essential Docker container management commands using an Nginx container as an example.

Understanding Docker Container Commands

Docker commands communicate with the Docker daemon, which manages Docker images, containers, networks, and volumes.

Some important container operations include:

- **List:** View running and stopped containers.
- **Stop:** Stop a running container.
- **Start:** Restart an existing stopped container.
- **Remove:** Delete a container permanently.

Example:

Suppose you have created an Nginx container to host a web application.

- You can use docker ps to check whether the container is running.
- You can use docker stop to stop the container.
- You can use docker start to start the same container again.
- You can use docker rm to delete the container.

Lab Steps

Step 1: View Running Containers

To display all currently running Docker containers, use:

```
docker ps
```

Explanation:

- Displays only running containers.
- Shows container ID, image name, command, creation time, status, ports, and container name.



Step 2: View All Containers

To display both running and stopped containers, use:

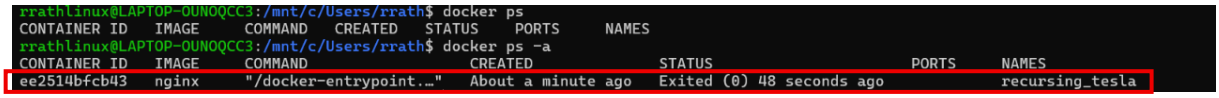
```
docker ps -a
```

Explanation:

- Displays all containers.
- Includes running containers.
- Includes stopped and exited containers.

Difference:

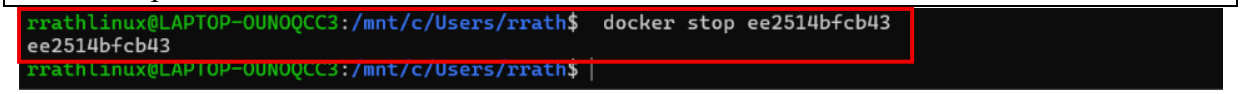
Command	Description
docker ps	Shows running containers only.
docker ps -a	Shows all containers, including stopped ones.



Step 3: Stop a Running Container

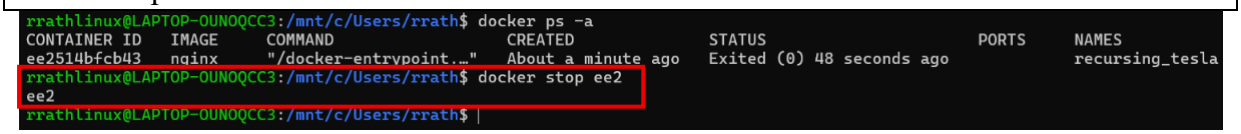
To stop a running container, use:

```
docker stop <container-id>
```



You can also stop a container using its name:

```
docker stop <container-name>
```



Example:

docker stop 344

Explanation:

- Stops the specified running container.
- You can provide the container ID or container name.
- After stopping the container, its application will no longer be accessible.

For example, if you are running Nginx on localhost, refreshing the browser after stopping the container will show that the website is unavailable.

Note: You can use the first few characters of a container ID instead of entering the complete ID, provided they uniquely identify the container.

Step 4: Start an Existing Container

To start an existing stopped container, use:

```
docker start <container-id>
```

```
rrathlinux@LAPTOP-OUNOQCC3:/mnt/c/Users/rrath$ docker start ee2
ee2
rrathlinux@LAPTOP-OUNOQCC3:/mnt/c/Users/rrath$
```

Example:

docker start 344

Explanation:

- Starts an existing stopped container.
- Uses the previously created container.
- Does not create a new container.
- Restores the container to its running state.

Step 5: Verify the Container Status

To check whether the container is running again, execute:

```
docker ps
```

You can also check all containers using:

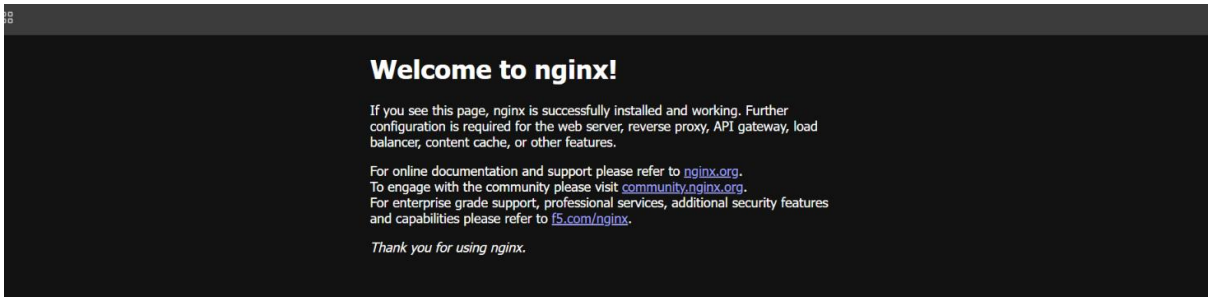
```
docker ps -a
```

Open your browser and navigate to:

```
rrathlinux@LAPTOP-OUNOQCC3:/mnt/c/Users/rrath$ docker ps
CONTAINER ID   IMAGE     COMMAND   CREATED   STATUS    PORTS     NAMES
rrathlinux@LAPTOP-OUNOQCC3:/mnt/c/Users/rrath$ docker ps -a
CONTAINER ID   IMAGE     COMMAND                  CREATED              STATUS                      PORTS     NAMES
ee2514bfcb43   nginx     "/docker-entrypoint.…"   About a minute ago   Exited (0) 48 seconds ago             recursing_tesla
```

```
http://localhost
```

If the Nginx container is running and port mapping is configured correctly, you should see the Nginx welcome page.



Step 6: Remove a Container

If you want to delete an existing container, first stop it and then remove it.

First, stop the container:

```
docker stop 344
```

Next, remove the container:

```
docker rm 344
```

```
rrathlinux@LAPTOP-OUNOQCC3:/mnt/c/Users/rrath$ docker stop ee2
ee2
rrathlinux@LAPTOP-OUNOQCC3:/mnt/c/Users/rrath$ docker rm ee2
ee2
rrathlinux@LAPTOP-OUNOQCC3:/mnt/c/Users/rrath$
```

Explanation:

- docker stop stops the running container.
- docker rm removes the container.
- Once removed, the container cannot be started again.
- You must create a new container using docker run if you need it again.

Step 7: Verify Container Deletion

To verify that the container has been deleted, execute:

```
docker ps -a
```

If the container was successfully removed, it will no longer appear in the list.

```
rrathlinux@LAPTOP-OUNOQCC3:/mnt/c/Users/rrath$ docker ps -a
CONTAINER ID   IMAGE     COMMAND   CREATED   STATUS    PORTS     NAMES
rrathlinux@LAPTOP-OUNOQCC3:/mnt/c/Users/rrath$
```

You can also verify the result in Docker Desktop:

1. Open Docker Desktop.
2. Navigate to **Containers**.
3. Check the container list.
4. The deleted container should no longer appear.